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UNIVERSIDAD DE EL SALVADOR

Facultad Multidisciplinaria De Occidente Departamento
de Ingeniería y Arquitectura Ingeniería en Desarrollo de
Software

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Asignatura: Matematica Aplicada 2

Tema: Parcial 3

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Ciclo II 2024 / Primer



⊕ Ejercicio 1.

$$x = \rho \cos \theta$$

$$y = \rho \sin \theta$$

$$z = z$$

$$x = 5 \cos \left(\pi/3 \right)$$

$$= 5 \times \frac{1}{2}$$

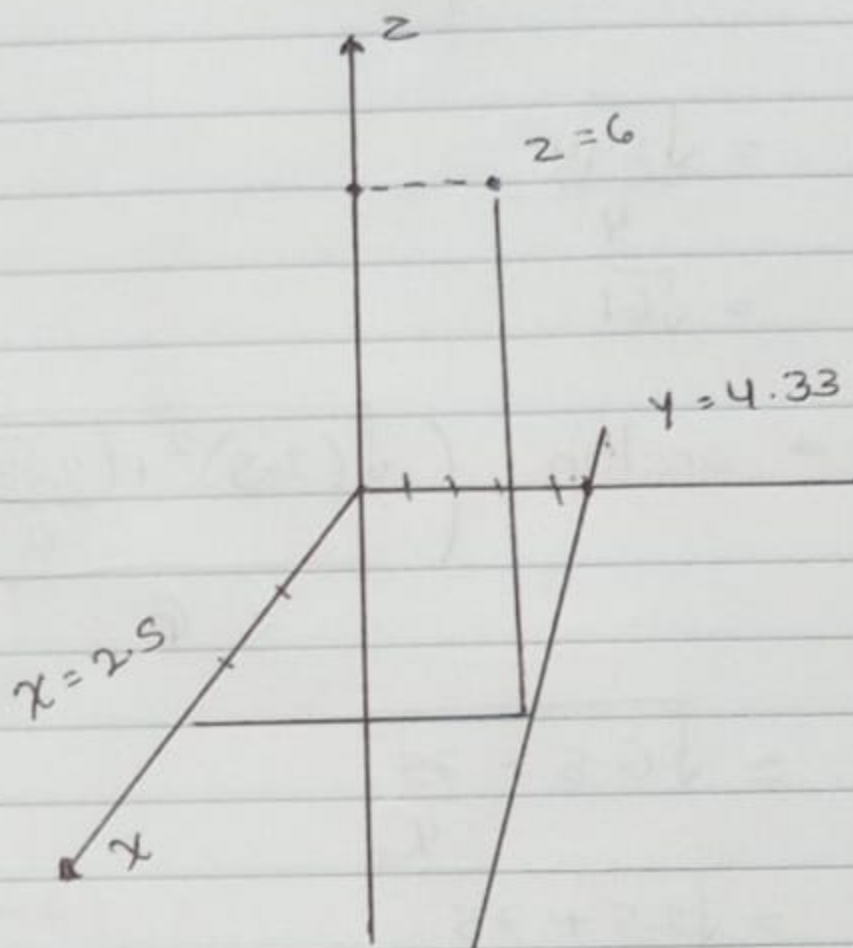
$$= 2.5$$

$$y = 5 \sin \left(\pi/3 \right)$$

$$= 5 \times \frac{\sqrt{3}}{2}$$

$$= \frac{5\sqrt{3}}{2}$$

$$z = 6$$



$$(x, y, z) = \left(2.5, \frac{5\sqrt{3}}{2}, 6 \right) \text{ Coordenadas cartesianas.}$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \arctan \left(\frac{\sqrt{x^2 + y^2}}{z} \right)$$

$$\theta = \arctan \left(\frac{y}{x} \right)$$

$$\rho = \sqrt{(2.5)^2 + \left(\frac{5\sqrt{3}}{2} \right)^2 + 6^2}$$

$$\rho = \sqrt{\frac{25}{4} + \frac{75}{4} + 36}$$

$$= \sqrt{\frac{20 + 75 + 144}{4}}$$

$$= \sqrt{\frac{244}{4}}$$

$$= \sqrt{61}$$

$$\phi = \arctan \left(\frac{\sqrt{(2.5)^2 + (2\sqrt{3})^2}}{6} \right)$$

$$= \sqrt{6.5 + \frac{75}{4}}$$

$$= \sqrt{\frac{25 + 75}{4}}$$

$$= \sqrt{\frac{100}{4}}$$

$$= 5$$

$$\phi = \arctan \left(\frac{5}{6} \right)$$

$$\theta = \arctan \left(\frac{5\sqrt{3}}{2} \right) = \arctan(\sqrt{3})$$

$$\frac{2.5}{2} = \pi/3$$

$$(p, \phi, \theta) = (\sqrt{61}, \arctan(\frac{5}{6}), \frac{\pi}{3})$$

$$(\sqrt{61}, 0.69, \pi/3)$$

⊕ Ejercicio 2

$$r = 4 \cos(2\theta)$$

$$A = \frac{1}{2} \int_a^b r^2 d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} 16 \cos^2(2\theta) d\theta$$

$$= 8 \int_0^{\pi/2} \cos^2(2\theta) d\theta$$

$$\cos^2(2\theta) = \frac{1 + \cos(4\theta)}{2}$$

$$= 8 \int_0^{\pi/2} \frac{1 + \cos(4\theta)}{2} d\theta$$

$$= 4 \int_0^{\pi/2} (1 + \cos(4\theta)) d\theta$$

$$1 \text{ Integral} = \int_0^{\pi/2} 1 d\theta = [\theta] \Big|_0^{\pi/2}$$

$$= \pi/2$$

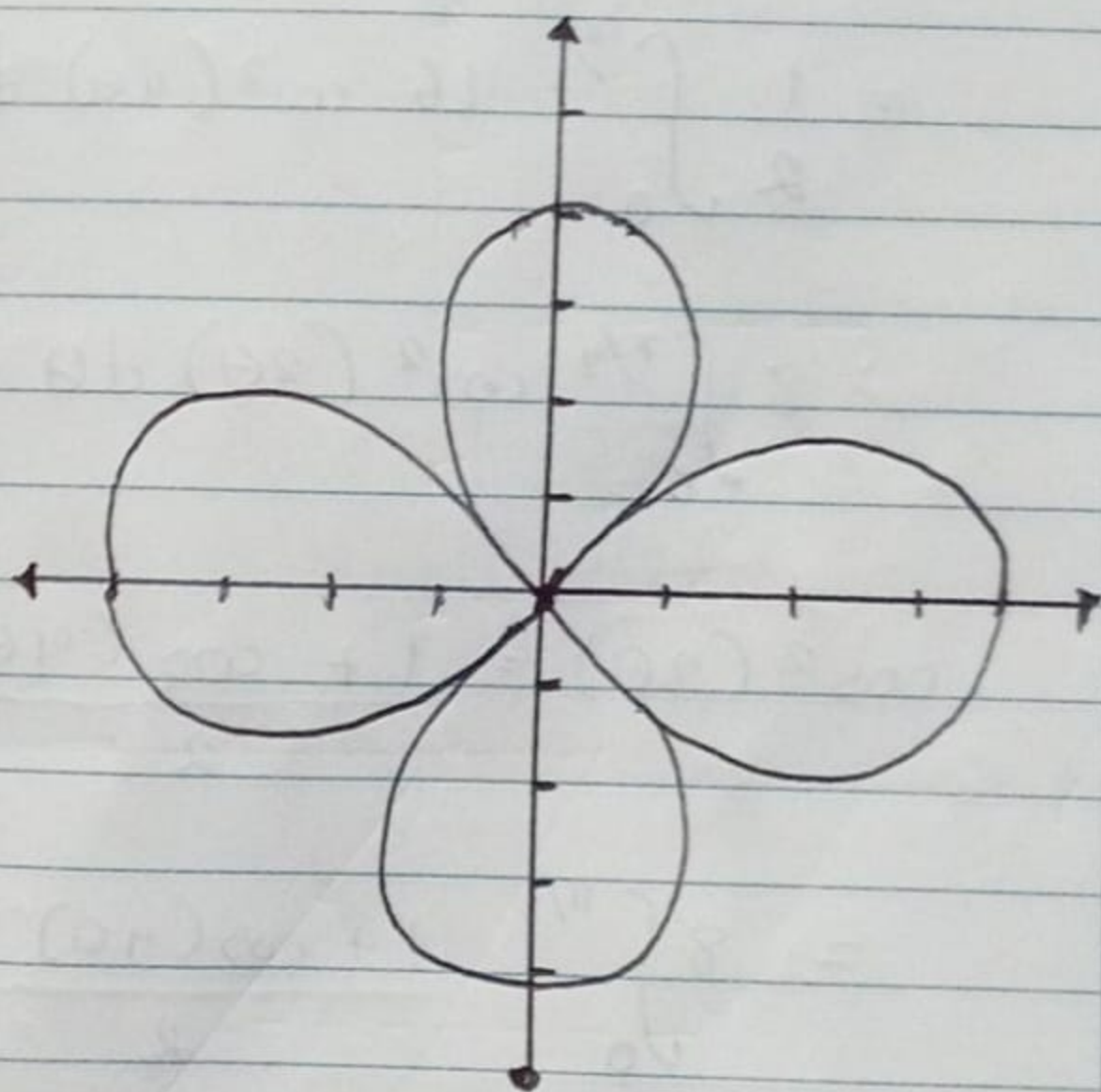
$$\text{II Integral} = \int_0^{\pi/2} (\cos(4\theta)) d\theta$$

$$= \frac{\sin 4\theta}{4} \Big|_0^{\pi/2}$$

$$= \frac{\sin(2\pi)}{4}$$

$$= \frac{\sin(\cancel{\theta})}{4}$$

$$= 0$$



$$A = 4 \left(\pi/2 + 0 \right)$$

$$= 4 \cdot \frac{\pi}{2}$$

$$= 2\pi$$

R// $r = 4 \cos(2\theta)$

$$A = 2\pi$$

⚡ Ejercicio 3

$$A = \int_{\theta_1}^{\theta_2} \int_{r_1}^{r_2} r \, dr \, d\theta$$

$$= \int_0^{2\pi} \int_0^3 r \, dr \, d\theta$$

$$= \int_0^3 r \, dr = \left. \frac{r^2}{2} \right|_0^3$$

$$= \frac{9}{2}$$

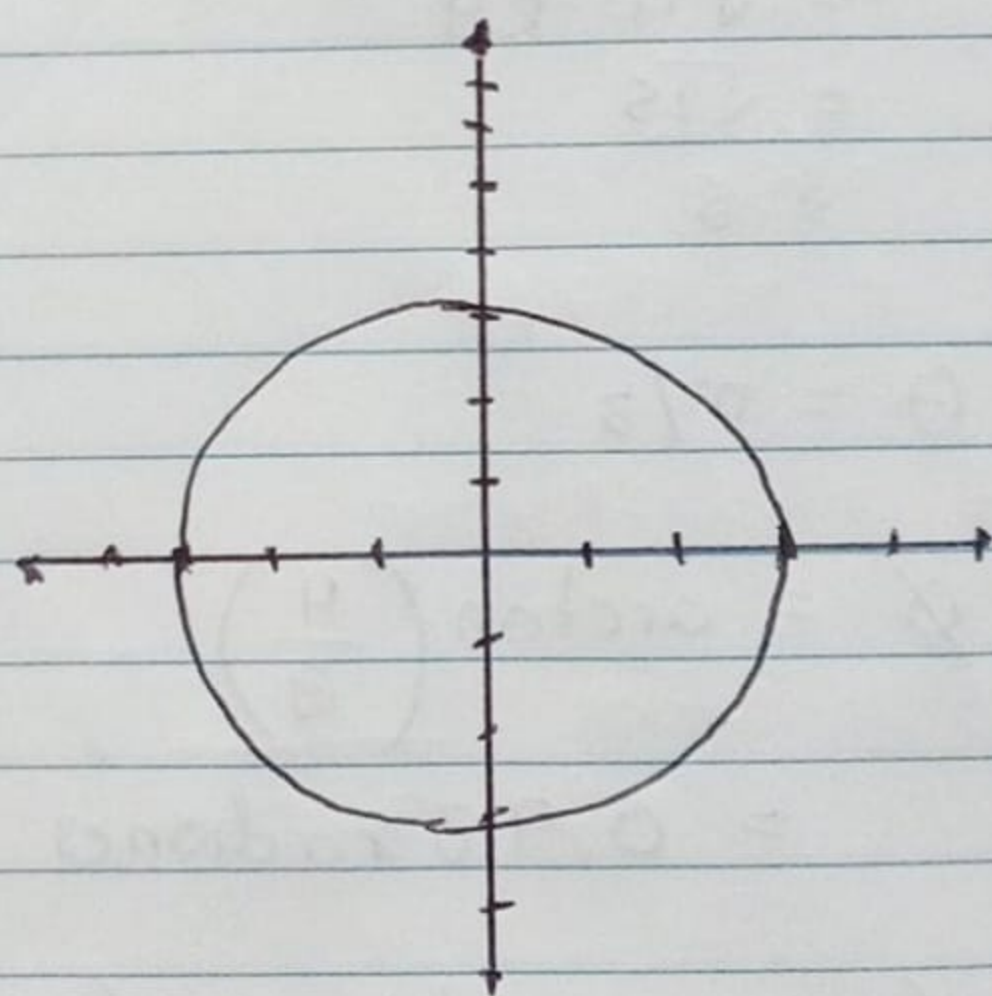
$$= \int_0^{2\pi} d\theta = \left. \theta \right|_0^{2\pi}$$

$$= 2\pi$$

$$A = \frac{9}{2} \times (2\pi - 0)$$

$$= 9\pi$$

$$A = 9\pi \approx 28.27$$



⊕ Ejercicio 4

$$\rho(\theta, \phi)$$

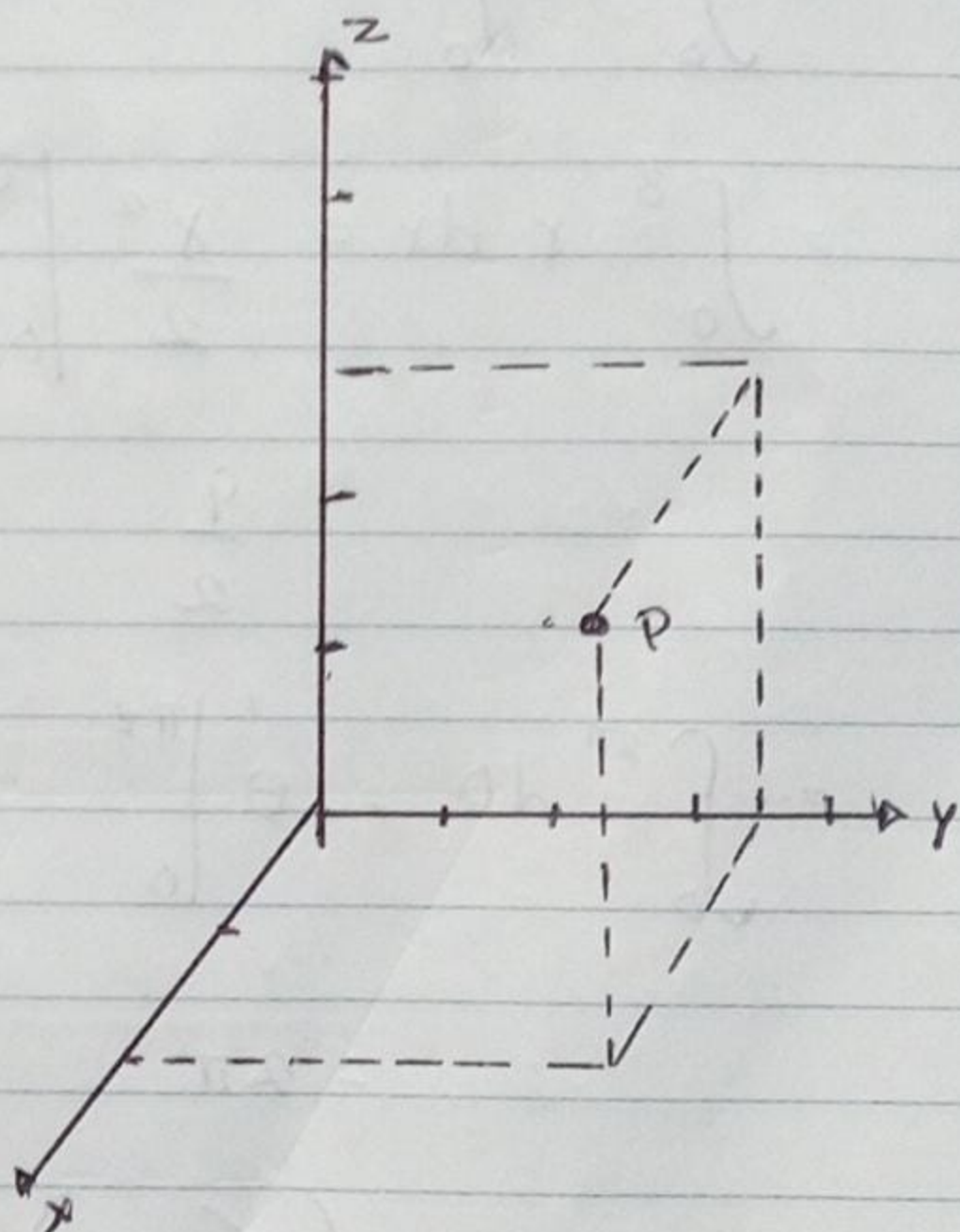
$$\rho = \sqrt{x^2 + z^2}$$

$$\phi = \arctan\left(\frac{y}{z}\right)$$

$$\begin{aligned}\rho &= \sqrt{4^2 + 3^2} \\ &= \sqrt{16 + 9} \\ &= \sqrt{25} \\ &= 5\end{aligned}$$

$$\theta = \pi/3$$

$$\begin{aligned}\phi &= \arctan\left(\frac{4}{3}\right) \\ &= 0.93 \text{ radianes}\end{aligned}$$



$$(\underset{\text{ó}}{\rho}, \theta, \phi) = (5, \pi/3, \arctan\left(\frac{4}{3}\right))$$

$$(\rho, \theta, \phi) = (5, \pi/3, 0.93)$$

Exercício 5

$$(x, y, z) = (3, -4, 5)$$

$$r = \sqrt{x^2 + y^2}$$

$$= \sqrt{3^2 + (-4)^2}$$

$$= \sqrt{9 + 16}$$

$$= \sqrt{25}$$

$$= 5$$

$$\theta = \arctan\left(\frac{y}{x}\right)$$

$$= \arctan\left(\frac{-4}{3}\right) + \pi$$

$$= -0.93 + \pi$$

$$= 2.21 \text{ rad}$$

$$z = 5$$

$$(r, \theta, z) = (5, 2.21 \text{ rad}, 5)$$

$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \arctan\left(\frac{y}{x}\right)$$

$$\phi = \arccos\left(\frac{z}{\rho}\right)$$

$$\rho = \sqrt{3^2 + (-4)^2 + 5^2}$$

$$= \sqrt{9 + 16 + 25}$$

$$= \sqrt{50}$$

$$= 7.071$$

$$\theta = 2.21 \text{ rad}$$

$$\phi = \arccos\left(\frac{5}{7.071}\right)$$

$$= \arccos(0.7071)$$

$$= 0.785 \text{ rad}$$

$$(\rho, \theta, \phi) = (7.071, 2.21 \text{ rad}, 0.785 \text{ rad})$$

